Nitish Kumar Sah

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Work Experience

Saptang Labs – Computer & Network Security **Dec 2025 – Apr 2026**
Software Engineer Intern | FastAPI, MongoDB, JWT, Redis, Celery, Docker Remote (Chennai) | Certificate

- Developed **50+ secure REST APIs** in Python using FastAPI with **authentication, RBAC, input validation, and activity logging**, enabling auditable and traceable access to critical application workflows.
- Designed a **multi-factor authentication system** with an **asynchronous email OTP pipeline** (API → Redis Queue → Celery Worker → SMTP), using **exponential retry handling (3 attempts)** to achieve **99%+ delivery reliability**.
- Optimized database-intensive API workflows using **indexing and Redis caching**, reducing average response latency by **30–40%** across the tested workload and query patterns.

Projects

NorthStar – Employee Management & Progress Tracking **May 2026**
Python, FastAPI, MongoDB, Redis, Celery, Docker, GitHub Actions, AWS EC2 GitHub | Live

- Built a **role-based goal management backend** using FastAPI and MongoDB with **25+ REST APIs**, **JWT authentication, RBAC** for Employee/Manager/Admin roles, and an **audit log** covering every goal action.
- Engineered a **5-state goal approval lifecycle** with **100% weightage and 8-goal validation**, **auto-scored quarterly check-ins** across **4 UoM types**, and **shared KPIs** that sync achievements to all assignees.
- Containerized **5 services (api, worker, mongo, redis, mailpit)** with **Docker Compose** and built a **GitHub Actions CI/CD pipeline** that blocks deploys on failing **pytest** tests and auto-deploys to **AWS EC2** on every push.

CampusSentinel – AI Campus Security System (Runner-up, Ethos 2025, IIT Guwahati) **Oct 2025**
Python, FastAPI, Django REST, MySQL, LangChain, LangGraph GitHub | Demo

- Designed a **microservice backend** in Python using **Django REST + FastAPI** with JWT authentication, load-tested at **200 requests/sec** while achieving a **p95 latency of 100ms** under sustained load.
- Modeled a **MySQL** database to index multi-source event logs, storing and correlating **10,000+ daily events** for continuous large-scale ingestion, retrieval, and analysis.
- Implemented an **agentic AI chatbot** with **FastAPI, LangGraph, LangChain, and Gemini**, enabling operators to query **6000+ tracked entities**, retrieve timelines, and triage alerts through a stateful **/chat** API endpoint.

Technical Skills

- **Languages:** Python, C++, SQL
- **Databases & Caching:** MySQL, MongoDB, Redis
- **Frameworks & Libraries:** FastAPI, Django REST, Pydantic, asyncio, pytest, React, Next.js
- **Tools & GenAI:** Git, GitHub Actions, JWT, Celery, Docker, AWS (EC2, RDS), LangChain, LangGraph
- **Concepts & Practices:** REST API Design, RBAC, JWT Authentication, Automated API Testing, Load Testing

Achievements/Certifications

- Achieved **Specialist** rank on Codeforces with a rating of **1400+** in ICPC-style contests. Codeforces
- Achieved a **3-Star** rating on CodeChef with a rating of **1600+**. CodeChef
- Led the team to win **Runners-Up** at Ethos 2025, a national hackathon by the Coding Club, IIT Guwahati. Certificate
- Secured **2nd Runners-Up** at Hackfest'25, a tech fest hackathon at IIT-ISM Dhanbad. Certificate
- Earned the **Advanced Learning Algorithms** certification from DeepLearning.AI. Certificate

Education

National Institute of Technology, Rourkela **Aug 2023 – May 2027**
B.Tech in Biotechnology *Rourkela, Odisha*

DAV Public School, Hehal, Ranchi **May 2022**
CBSE, Science (PCM) *Ranchi, Jharkhand*

Relevant Coursework

- Data Structures & Algorithms
- Object Oriented Programming
- Database Management Systems
- Operating Systems
- Computer Networks
- Deep Learning

Extracurricular Activities

- **Pantomime** | **Dramatics Club** | Treasurer | Mar 2024 – Present
- **Webwiz** | **Web Development Club** | Technical Team Member | Jan 2025 – Present